

1.2.4 – 1.2.7 Waterfall vs Agile Models & Case Studies

وائرفال (آبشار) ماڈل ایک سیدھا اور ترتیب وار طریقہ ہے؛ ہر مرحلہ مکمل ہونے کے بعد ہی اگلا شروع ہوتا ہے اور پیچھے جانا مشکل ہوتا ہے۔

CS XI

Chapter 1: Computer Systems

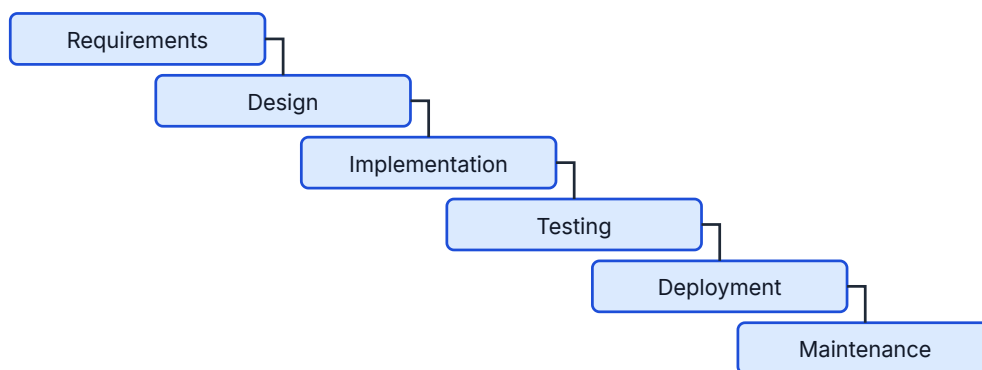
Lecture 10 of 12

★ Golden

1.2.4 – 1.2.7 Waterfall vs Agile Models & Case Studies ★ Golden

LEARN IT

The **Waterfall model** is a **linear, sequential** approach to the SDLC. Each phase must be completed before the next one begins, like water falling down steps. There is no easy way to go back to a previous phase.



The Waterfall model: each phase "falls" into the next, in one direction only.

When to use it

- Requirements are clear, fixed and well understood.
- The project is small to medium in size.
- The technology is stable (not new or experimental).

Advantages (Pros)	Disadvantages (Cons)
Simple to understand and manage.	Not flexible to changes.
Well documented at every step.	Testing happens very late in the process.

WORKED EXAMPLE

Case study: Al-Noor Library Management System. The textbook uses this system as a Waterfall case study because the requirements were **100% fixed before coding began**.

Step 1: The librarian listed exactly what was needed: add books, issue books, return books, calculate fines. Nothing would change.

Step 2: Because the list was fixed, the team could finish the full design, then all the coding, then all the testing, in order.

Step 3: Every phase produced a document, so the library has a complete record for future maintenance.

REMEMBER IT

Waterfall = water only flows downwards. Once water has fallen it cannot climb back up — once a phase is finished you do not go back.

COMMON MISTAKE

Do not choose Waterfall for a project where the customer is still unsure what they want. A late change means going back through design, coding and testing, which is slow and costly.

اردو میں سمجھیں

وائرفال (آبشار) ماڈل ایک سیدھا اور ترتیب وار طریقہ ہے؛ ہر مرحلہ مکمل ہونے کے بعد ہی اگلا شروع ہوتا ہے اور پیچھے جانا مشکل ہوتا ہے۔ یہ تب بہتر ہے جب ضروریات واضح اور طے شدہ ہوں، جیسے النور لائبریری مینجمنٹ سسٹم میں۔ یہ سمجھنے میں آسان ہے مگر تبدیلیوں کے لیے لچکدار نہیں اور ٹیسٹنگ دیر سے ہوتی ہے۔

EXAM TIP

Expect "Explain the Waterfall model with its advantages and disadvantages" or "When should the Waterfall model be used?" Draw the step diagram, list the three "when to use" conditions, and give at least two pros and two cons. Mention the Al-Noor Library case study as your example.

CHECK YOURSELF

1. Why is the Waterfall model called "sequential"?
2. Why did the Al-Noor Library system use Waterfall?

Answers: 1. Because each phase must be completed before the next begins, in a fixed order. 2. Because its requirements were 100% fixed before coding began.

LEARN IT

The **Agile model** is an **iterative** approach: the software is built in small, repeated cycles called **sprints** (usually 2–4 weeks long). Each sprint plans, builds and tests a small part of the software, and then shows it to the customer. **Customer feedback is gathered continuously** and used in the next sprint.



When to use it

- Requirements are unclear or may change.
- The customer wants frequent progress updates (early delivery).
- The project is complex or innovative.

Advantages (Pros)	Disadvantages (Cons)
Highly flexible to changes.	Less documentation.
Early delivery of working software.	Unpredictable end-time and cost.

WORKED EXAMPLE

Case study: Al-Noor Student Mobile Application. This app used Agile because teachers, parents and students **constantly requested new features** during development.

Sprint 1 (2 weeks): build login and the timetable screen. Show it to users.

Feedback: parents ask, "Can we see attendance too?"

Sprint 2: add attendance. Teachers now ask for homework notifications.

Sprint 3: add notifications. Each sprint delivers a working app, and new ideas are welcomed instead of causing a restart.

REMEMBER IT

Agile = agile athlete: quick and able to change direction. Think of making a pizza where the customer tastes each slice and you adjust the next one.

COMMON MISTAKE

"Less documentation" does not mean "no documentation" or "no planning". Agile teams still plan every sprint; they simply value working software over long documents.

اردو میں سمجھیں

ایجائل (Agile) ماڈل میں سافٹ ویئر چھوٹے چھوٹے ادوار میں بنتا ہے جنہیں اسپرنٹ (Sprint) کہتے ہیں، جو عموماً دو سے چار ہفتے کے ہوتے ہیں۔ ہر اسپرنٹ کے بعد گاہک کی رائے لی جاتی ہے۔ یہ تبدیلیوں کے لیے بہت لچکدار ہے اور جلد کام کرنے والا سافٹ ویئر دیتا ہے، مگر اس میں دستاویزات کم ہوتی ہیں اور وقت و لاگت کا اندازہ مشکل ہوتا ہے۔ انور اسٹوڈنٹ موبائل ایپ اس کی مثال ہے۔

EXAM TIP

Questions ask you to define Agile, explain sprints, list pros/cons, or say when to use it. Always mention the words "iterative", "sprints (2–4 weeks)" and "continuous customer feedback". Use the AI-Noor Student Mobile App as your case study.

CHECK YOURSELF

1. What is a sprint and how long does it usually last?
2. Give one disadvantage of Agile.

Answers: 1. A short development cycle that delivers a working part of the software; usually 2–4 weeks. 2. Less documentation, or unpredictable end-time and cost.

LEARN IT

Both models follow the same SDLC phases. The difference is **how** they move through them: Waterfall goes through each phase **once**, in a straight line; Agile goes through all phases **many times**, in small loops.

Dimension	Waterfall model	Agile model
Flexibility	Rigid. Hard to make changes later.	Very flexible. Adapts to changes easily in the next sprint.
Customer involvement	Only at the start (requirements) and the end (delivery).	Continuous throughout the project (after every sprint).
Delivery of software	Single delivery at the very end.	Frequent, partial deliveries after every sprint.
Testing	Done late, in a dedicated testing phase.	Continuous testing within every sprint.
Documentation	Heavy documentation required.	Light documentation. Focus is on working software.

WORKED EXAMPLE

Choosing a model — think like the textbook.

Case A: Al-Noor Library Management System. Requirements fixed 100% before coding → no changes expected → **Waterfall**.

Case B: Al-Noor Student Mobile Application. Teachers, parents and students keep asking for features → requirements change → **Agile**.

Practice case: A bank wants a salary-calculation program for a fixed government formula that will not change. Requirements are clear and fixed, the project is medium-sized, and the technology is stable → **Waterfall**.

Practice case: A start-up wants a new food-delivery app and is not sure which features customers will like. Requirements are unclear and the project is innovative → **Agile**.

REMEMBER IT

One question decides it: "**Will the requirements change?**" No → Waterfall. Yes / not sure → Agile.

COMMON MISTAKE

Do not say one model is always "better". Each suits a different situation. In exams, justify your choice using the project's features (fixed vs changing requirements, size, customer involvement).

اردو میں سمجھیں

وائرفال ماڈل سخت (Rigid) ہے، گاہک صرف شروع اور آخر میں شامل ہوتا ہے، سافٹ ویئر ایک ہی بار آخر میں ملتا ہے اور دستاویزات زیادہ ہوتی ہیں۔ ایجائل ماڈل لچکدار ہے، گاہک ہر اسپرنٹ کے بعد شامل ہوتا ہے، سافٹ ویئر بار بار حصوں میں ملتا ہے اور ٹیسٹنگ مسلسل ہوتی ہے۔ اگر ضروریات طے شدہ ہوں تو وائرفال اور اگر بدلتی رہیں تو ایجائل منتخب کریں۔

CHECK YOURSELF

1. In which model is the customer involved only at the start and the end?
2. Which model delivers software in frequent, partial deliveries?
3. A school wants an exam-result system with rules that never change. Which model?

Answers: 1. Waterfall. 2. Agile. 3. Waterfall (fixed, clear requirements).

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Waterfall is:

- a. Linear — finish a phase before the next
- b. 2-week sprints
- c. Random order
- d. A gate

MCQ 2. Agile builds software in:

- a. Short sprints with feedback
- b. One giant leap only
- c. Hardware NAND
- d. OSI layer 1

MCQ 3. You cannot easily go back in:

- a. Waterfall
- b. Agile
- c. Both equally
- d. Neither

MCQ 4. Customer sees a working slice first in:

- a. Agile
- b. Pure waterfall
- c. K-maps
- d. Binary search

MCQ 5. A hospital records system with frozen law rules fits:

- a. Waterfall
- b. Only Agile memes
- c. No SDLC
- d. HCI pie charts

MCQ 6. Changing shop-app features weekly fits:

- a. Agile
- b. Waterfall only
- c. TCP
- d. Selection sort

Extra short questions

1. Define Waterfall and Agile in one line each.
2. Give one advantage of each.
3. Give one disadvantage of each.
4. When would you pick Waterfall?

Extra long question

1. Compare Waterfall vs Agile on five dimensions and attach one Sindh-college case to each model.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Waterfall: sequential phases, no going back. Agile: iterative sprints (about 2–4 weeks) with customer feedback.
2. Waterfall: clear documents and deadlines. Agile: change is cheap and the user sees value early.
3. Waterfall: late change is painful. Agile: needs an available customer and can look undocumented.
4. Stable, legally frozen requirements (e.g. a regulated records system).

Long

1. Dimensions: order, change, customer, documentation, risk. Case Waterfall: board exam-result system with fixed rules.
Case Agile: college app whose features keep changing after student feedback.